

Prediction of Chronic Obstructive Pulmonary Disease using CNN-GRU model

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Abstract:

COPD is a lung disease that gets worse over time. It causes you to have trouble breathing (dyspnea), wheeze, and cough all the time. Because of the vital requirement for early identification and care in respiratory illnesses, audio analysis of breathing sounds has emerged as an important technique in the prediction of COPD. This research uses deep learning approaches, especially a hybrid model that combines CNN and GRU techniques can be applied to increase the accuracy of COPD diagnosis. The algorithm has been trained using a dataset that includes breathing sounds from COPD patients and healthy people. Feature extraction from audio data is used to identify important properties, which are subsequently processed using the CNN-GRU model for classification purposes. Integration of CNN for spatial feature extraction with GRU. Temporal sequence learning has showed great potential in distinguishing between COPD and healthy persons. The findings show that this deep learning technique is very accurate and has the potential for cost-effective, Continuous remote monitoring for medical applications, assisting in the early detection and management of COPD exacerbations.

Keywords: COPD, CNN, GRU, MFCCs, Respiratory Illnesses, Deep Learning

I. Introduction:

In recent years, lung disease has become the third leading cause of death in the world. World Health Organization (WHO) estimates show that the top five lung diseases are lung cancer, chronic obstructive pulmonary disease (COPD), tuberculosis and Acute Lower Respiratory Tract Infection (ALRTI)[1, 2]. Obstructive diseases are those in which the constriction of airways results in a decrease in the amount of air exhaled from the lungs[3].

COPD, or chronic obstructive pulmonary disease, refers to the presence of emphysema, chronic bronchitis, and/or intractable asthma. Diseases of this family of lung diseases that limit the passage of air to the alveoli are dangerous for both industrialized and developing countries. In low-income countries, the mortality of about 90% of the population due to this disease is reported. The Global Initiative for Chronic Obstructive Lung Disease (GOLD) results reveal that COPD was the 3rd leading cause of death in 2020[4]. Despite decades of research on COPD, the medical community has not made substantial headway in lowering morbidity and mortality. Other significant noncommunicable diseases One key element contributing to this delayed recovery might be because prior COPD definitions revealed flaws, such as the absence of diagnosing the condition in its early stages. In the absence of flow constraints[5]. Lung illnesses, particularly Chronic Obstructive Pulmonary Disorder (COPD), are a major worldwide health problem. Accurate and early identification of COPD and associated disorders is critical, but it is sometimes impeded by the complexities of medical imaging and the variability in disease presentations[6]. Chronic obstructive pulmonary diseases can be diagnosed in a variety of clinical approaches. Chest X-rays, CT scans, and MRIs are some of the most often utilized imaging procedures for diagnosis[7, 8]. However,

employing these imaging modalities has significant obstacles, such as[1, 9]:

- The frequent use of imaging procedures such as X-rays and CT scans exposes patients to ionizing radiation[10], which can raise the risk of health problems such as cellular damage or cancer over time. This is especially important for individuals who need long-term monitoring.
- Advanced imaging technologies, such as CT and MRI scanners, are expensive to acquire and maintain. The necessity for specialist staff drives up costs, restricting access to these technologies, particularly in resource-constrained healthcare settings.
- Remote places frequently lack the infrastructure and resources required for modern imaging technology. Transporting patients to metropolitan centers is difficult, and establishing machines in these areas is sometimes impossible owing to logistical and technological difficulties.

This work aims to test the effectiveness of the deep learning model regarding traditional diagnostics currently used in the medical field for accurate COPD prediction, Early detection and close monitoring of respiratory infections are critical for effective therapy.

The sounds made by the lung structures when breathing are known as breath sounds. It is most effective to use a stethoscope to hear the lung sounds. The clinician may detect reduced or non-existent breathing sounds, aberrant breathing sounds, and regular breathing sounds with a stethoscope[12]. A common clinical procedure for diagnosing breathing disorders is draping a stethoscope over a patient's chest. These sounds are usually classified as normal or abnormal. The most familiar odd sounds heard above normal lung sounds are crackles, wheezes and squeaks which usually indicate a lung disease. Normal lung sound is often cyclical and represents the passage of air into and

out of lungs while breathing. Pulmonary disease refers to chronic bronchitis and emphysema, where there's blockage of air flow in the lung which hinders normal breathing. And this blockage is not fully reversible (it cannot be completely cured). Using a stethoscope obviously refers to a qualitative diagnostic process that provides quite direct and objective information.[13].

To obtain correct prediction of COPD, the combination of CNN and GRU takes the advantages of both the architecture. CNNs are extremely good at extracting spatial features from input data, such as identifying patterns in spectrograms or MFCCs (Mel-Frequency Cepstral Coefficients) in audio signals, which can indicate lung issues. GRUs, on the other hand, are designed to handle sequential input and capture temporal dependencies, making them perfect for handling time-series data, such lung sounds' breathing cycles. By merging these two models, the architecture can discover detailed patterns in the input and trace the advancement of these patterns over time, improving its ability to distinguish between healthy and COPD-affected individuals.

II. Related work:

Image databases have been used in a number of studies on the prediction of chronic obstructive pulmonary disease. The severity of asthma and COPD exacerbations was assessed using machine learning techniques. The classification of COPD in CT scans was done using a 3D convolutional neural network.[14]. Lung disease has been diagnosed using Convolutional Neural Networks and Mel-frequency cepstral coefficients of respiratory sounds.[15, 16]. Funda and others used SVMs and CNN to detect respiratory sounds. The study achieved an 83% classification accuracy using VGG16-CNN-SVM deep learning models[17]. Arka et al[18] used diverse lung sound collection to develop RDLINet, a deep learning model for classifying respiratory illnesses. In order to improve model performance, R. Du et al.,[19] emphasized on representing airway abnormalities well in COPD diagnosis. A deep CNN model is constructed to examine 3D lung airway trees from CT image snapshots. The model is optimized using Bayesian algorithms to improve classification performance. Class-discriminative localization maps are produced using Grad-CAM, enhancing interpretability and decision clarity.

Materials and Methods

A. Dataset

The dataset consists of 3,541 lung sound recordings from persons who are healthy and patients with COPD. To make sure the model is trained on representative data, these audio samples—which come from a variety of medical institutions—cover a wide spectrum of respiratory disorders. Every recording, which lasts anywhere from thirty to ninety seconds, is made in a controlled setting to reduce ambient noise and is stored in WAV format for high-fidelity audio recording. The patient population

is made up of adults, seniors, and children.[1]. Labelled metadata is added to the dataset to provide more critical details about the patient characteristics and particular lung diseases linked to each recording. This extensive dataset makes it easier to create reliable machine learning models that can discriminate between lung sounds that are healthy and those that are impacted by COPD.

B. Implementation tools used

The model was implemented in Google Colab using the Python programming language. The Librosa package was used to extract features from the audio data, and the Matplotlib library was used to visualize the data. Convolutional neural networks (CNNs) were implemented using Keras, a deep learning framework based on TensorFlow. Additionally, TensorFlow was utilized to train the model and manage the underlying calculations. This set of tools and libraries enabled the successful development and implementation of the CNN-GRU architecture for assessing lung sounds and forecasting Chronic Obstructive Pulmonary Disease (COPD).

C. Data preprocessing

To guarantee that lung sound recordings are suitably prepped for the deep learning model, data pretreatment is a crucial step. Using the librosa package, the model concentrated on obtaining Mel-Frequency Cepstral Coefficients (MFCC) from every audio file. MFCCs are well-suited for capturing the spectral characteristics of audio signals, and in the context of lung sounds[15, 16, 20], they help in isolating key features such as breathing patterns, crackles, or wheezes, which are critical for detecting respiratory conditions. By transforming the raw audio into MFCCs, we convert the time-domain lung sound signals into a set of frequency-based features that offer a more robust representation for machine learning models.

A fixed-size input vector for every audio file was created by averaging the MFCC features over time frames after they were retrieved. This is essential for reliable model training. Regardless of the duration of the original recordings, this guarantees that all incoming data has the same dimensionality. The data was separated into training and testing sets after this feature extraction, enabling a thorough and objective assessment of the model's capacity to forecast respiratory diseases from unseen data. By dividing the dataset, the model's performance on the training data is evaluated, but so is its ability to generalize to new, untested samples. With its focus on MFCC extraction, this preprocessing pipeline provides a solid basis for precise and significant lung sound classification.

```

y_train
array([[1., 0.],
       [0., 1.],
       [1., 0.],
       ...,
       [1., 0.],
       [1., 0.],
       [0., 1.]])

```

Figure 1. Labels Encoded One-Hot

D. Models based on deep learning

Convolutional Neural Network

Convolutional Neural Networks have become a potent tool for audio dataset analysis, especially for uses such as lung sound classification. CNNs for audio work with audio waveforms, which are typically converted into spectrograms or Mel-spectrograms that track the frequency content of sound over time[20]. Techniques such as Short-Time Fourier Transform (STFT) and Mel-Frequency Cepstral Coefficients (MFCCs) are employed for audio conversion. Convolutional Neural Networks (CNNs) play a crucial role in extracting important features from spectrograms using their convolutional layers. These filters are designed to identify patterns and changes over time, while pooling layers help reduce the data size, improving processing efficiency. CNNs have shown great success in a range of audio classification tasks.

Gated Recurrent Unit(GRU):

GRUs are particularly effective for processing sequential audio signals because they maintain a hidden state that captures temporal relationships, which is advantageous for NLP and time-series tasks. Their gating mechanisms control the flow of information, enabling them to retain relevant historical data while filtering out less important details. This ability is crucial for audio applications, where classifying sound events heavily depends on timing and context. When GRUs are paired with CNNs in analyzing lung sounds, the model's capacity to learn both spectrogram features and temporal dynamics is improved, resulting in better accuracy in differentiating between healthy and unhealthy lung sounds.

E. CNN and GRU:

To efficiently evaluate audio input, the model uses Gated Recurrent Units (GRU) and Convolutional Neural Networks (CNN). The model's input shape, which represents the recovered Mel-frequency cepstral coefficients from the audio files, is set to (40, 1). Two 1D convolutional layers are placed after a Reshape layer in the design, which prepares the input for the CNN. A Batch Normalization layer and a MaxPooling layer that lowers dimensionality come after the first convolutional layer, which has a kernel size of 2 and employs 32

filters. Following Batch Normalization and MaxPooling, the second convolutional layer makes use of 64 filters with a kernel size of 2. Before being sent to the GRU layers, the output of the last MaxPooling layer is compressed and rearranged to preserve the sequential information. Two GRU layers are included in the model; the first one is used to return sequences for temporal processing, which improves the network's capacity to identify long-term dependencies in the audio feed. Dropout layers with a rate of 0.3 are added after each GRU layer to reduce overfitting. The model concludes with a two-unit dense output layer that provides class probabilities for healthy and ill lung sounds using a SoftMax activation function. By combining the capabilities of recurrent layers for temporal analysis and convolutional layers for feature extraction, this CNN-GRU combo improves classification accuracy for intricate audio patterns.

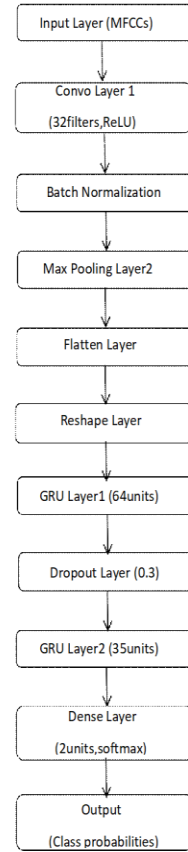


Figure 2. Architecture of CNN-GRU

III.

Performance evaluation

Evaluation metrics are quantitative measures that are used to evaluate the effectiveness of a system, approach, or model.

A. Accuracy

A crucial parameter in classification is accuracy, which quantifies the percentage of labels that are properly predicted. It offers a broad picture of performance, but for imbalanced datasets, it could require additional measurements.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Fig. 3 displays the CNN + GRU model's training and validation accuracy.

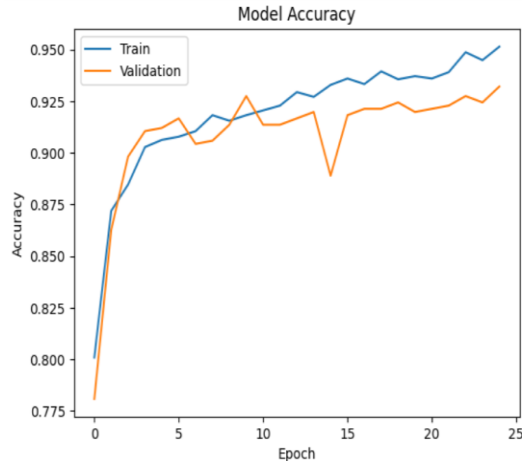


Figure 3. Accuracy of CNN-GRU model training and validation, revealing consistent improvement by epochs and significant generalization towards unseen data.

Fig. 4 displays the CNN + GRU model's training and validation loss.

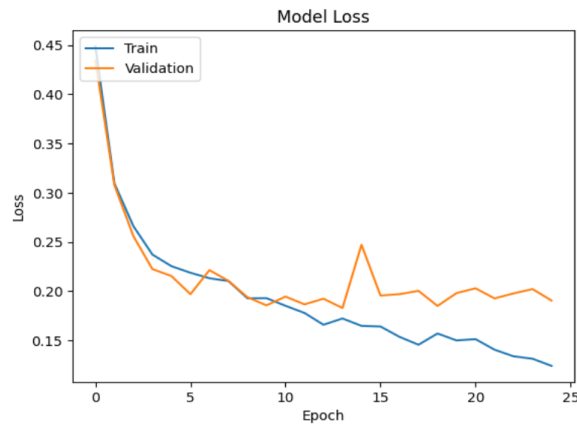


Figure 4. Loss of Training and Validation

In this case, the CNN-GRU model's accuracy is 93%.

B. Precision

In order to determine how many positive predictions are truly accurate, precision calculates the percentage of true positives among all anticipated positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

The number of instances in which the model accurately predicted a positive outcome is represented by True Positive (TP) in this instance.

False Positives (FP) are instances in which the model mis predicted a positive result.

C. Recall

The model's recall indicates how well it can detect positive occurrences by calculating the proportion of genuine positives among all actual positives. When the cost of missing positive instances is significant, it becomes even more crucial.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Number of instances in which the model accurately predicted a positive outcome is represented by True Positive (TP) in this instance.

The number of instances in which the model mispredicted a negative outcome is represented by False Negatives (FN).

D. F1 score

F1-score is commonly used to analyze classification models since it delivers a fair assessment of a model's performance. This harmonic mean is especially useful in situations when there is an unequal distribution of classes and emphasizes the need of accurately identifying positive cases and minimizing false positives.

$$\text{F1 score} = 2 * \frac{\text{Precision} * \text{recall}}{\text{Precision} + \text{recall}}$$

Figure 5. The accuracy, F1-score, and COPD support results are more promising.

21/21 ————— 0s 15ms/step
Precision: 92.40%
Recall: 92.59%
F1 Score: 92.44%

Classification Report:				
	precision	recall	f1-score	support
Non-COPD	0.94	0.97	0.95	525
COPD	0.84	0.76	0.79	123
accuracy			0.93	648
macro avg	0.89	0.86	0.87	648
weighted avg	0.92	0.93	0.92	648

Figure 5. Precision, Recall, F1-score measures for COPD classification, showing high precision rate, minimizing false positives, and good recall, reducing missed COPD cases

E. Confusion matrix

The confusion matrix is often displayed as a table with the actual labels in the rows and the expected labels in the columns. The cases that were successfully classified are shown on the table's diagonal, whereas the examples that were incorrectly classified are shown in the off-diagonal areas.

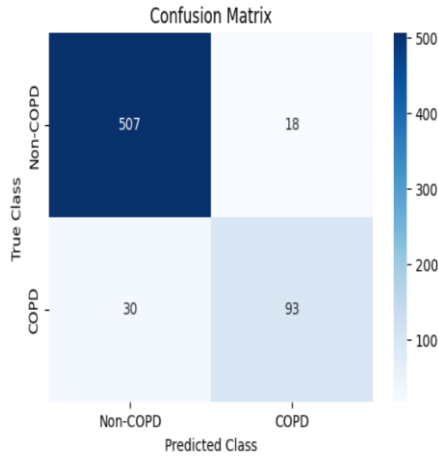


Figure 6. Confusion Matrix

F. Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1-Score
CNN + LSTM	92.0	90.3	91.1	90.7
CNN + GRU (Proposed)	93.0	91.5	92.3	91.9

G. ROC Curve:

The ROC curve evaluates a classifier by plotting the True Positive Rate (y-axis) against the False Positive Rate (x-axis) at various thresholds. It illustrates the trade-off between specificity and sensitivity; higher performance is indicated by a curve nearer the top-left. This is summed up by the AUC value, where stronger classifiers are indicated by higher values.

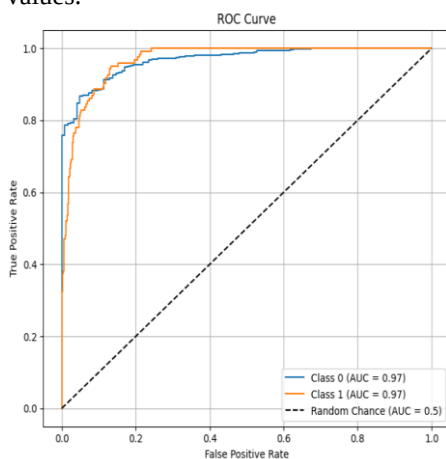


Figure 7. ROC Curve

H. Feature Distribution Histogram

By visualizing the frequency of data points for a certain feature, a feature distribution histogram shows how the

feature is distributed across several datasets or categories (e.g., training vs. testing). Histogram overlaying facilitates distribution comparison, imbalance detection, and consistency assurance.

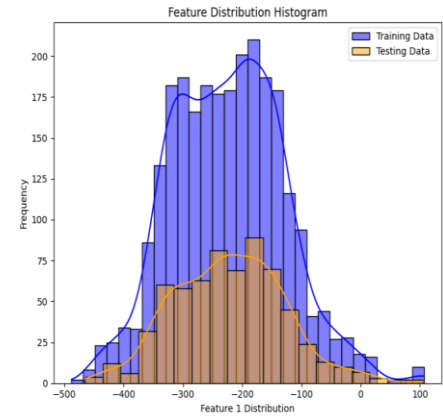


Figure 8. Feature Distribution Histogram

IV. Conclusion

To sum up, breathing sound analyses using deep learning models have shown tremendous accuracy in detecting chronic obstructive pulmonary disease (COPD)[1, 7]. sophisticated techniques like gated recurrent units (GRUs) in conjunction with convolutional neural networks (CNNs), have enabled more exact classification of respiratory diseases. A number of researchers looked into image-based methods. Nonetheless, it is novel and intriguing to simulate sounds with deep convolutional neural networks (CNN) and a gated recurrent unit (GRU).

Combining Gated Recurrent Units (GRUs) with Convolutional Neural Networks (CNNs) is an effective method for evaluating respiratory sounds and detecting chronic obstructive pulmonary disease (COPD). This hybrid model improves classification accuracy by using CNNs to extract spatial data from lung sound spectra and GRUs to capture temporal relationships. The capacity to interpret sequential data enables a more detailed knowledge of respiratory patterns, resulting in more accurate diagnosis of COPD problems.

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